

Penalty necessity on every (M0)–(M2) mesh, a global mesh containing the star counterexample, and tall boundary triangles

notes/v7/penalty

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Summary

Notation as in `notes/v6/consts/REPORT.tex`: $A(v) = \int_{\Omega_h} \frac{1}{2} Dv : Dv$, $B(v) = \langle \partial_n v, v \rangle_{\Gamma_h}$ (n the outward normal of the fluid), $C(v) = \|v\|_{\Gamma_h}^2$, $N_h(v, v) = A - 2B + (\mu/h)C$, μ^* the coercivity threshold of N_h on Z_h , θ the minimum angle, c_0 the constant of (M0).

- (1) **Goal 1: penalty necessity with no local quasi-uniformity [PROVED]**. Theorem 2: there are $c_*(\theta, c_0) > 0$ and $h_*(\theta, c_0, L) > 0$ such that on *every* conforming shape-regular mesh with $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ ($e \in \mathcal{E}_\Gamma$) and $h \leq h_*$, N_h is indefinite on Z_h (every $k \geq 4$) whenever $\mu < c_* \rho$. (M1), (M2), star or layer structure, and local quasi-uniformity are *not* needed. The same proof gives Clough–Tocher, $k \geq 2$, **[PROVED modulo the standard HCT interpolation estimate]**. The key is a witness whose stream function is *exactly* a quadratic in the normal variable on every boundary triangle. Its non-polynomial part lives only in two transition bands far from Γ_h . There the $O(1)$ -relative interpolation error that defeated the v6 approach costs only a small fraction of the gain. The constants are explicit but astronomically small ($c_*/c_0 \approx 10^{-39}$ at $\theta = 30^\circ$). The theorem is a statement about the scaling $\mu^* \gtrsim \rho$, not a usable threshold.
- (2) **Goal 2: a global mesh realising Proposition pb:counter [NUMERICAL; mesh hypotheses checked]**.
 - The mesh is a polar column layer of height $H\ell$ (then dyadic rings, then a structured outer part), with $N = 128, 256$ and all chords equal. Checked: (M0) with $c_0 = 1$, (M1) with no singular vertex, (M2) with $\Theta_0 \geq 0.77$, and minimum angle 11.25° ($H = 5$).
 - The boundary star carries no witness. Unions of 2, 3, 4, ... consecutive stars do (patch thresholds 0.43, 1.11, 1.46, ...).
 - The global threshold is $\gamma^* = \mu^* h_\Gamma / h = 3.056$ ($N = 128$) and 3.128 ($N = 256$). It is identical to 4 digits for $L = 2.5, 5, 10$, while ρ runs over 21.6–416. So $\mu^* = \gamma^* \rho$ exactly scales like ρ : the counterexample is local, not global.
 - $H = 8$ gives the same picture: no witness on 1–3 stars, and $\gamma^* = 1.853$ for every L .

Theorem 2 proves the scaling for this family as $h \rightarrow 0$.

- (3) **Goal 3: tall boundary triangles on coarse meshes.**

- **[PROVED, exact rational]** For stacked column meshes of *any* height H (boundary triangles right-angled with apex $\arctan(1/H)$, which is exactly the class where $\lambda_T < 0$ and the stars fail), a separable witness lying in the discrete space gives $\mu^* \geq (0.85/H) h/h_\Gamma$ at every mesh size (Proposition 6).

- **[NUMERICAL]** On the polar column meshes, $\gamma^* \cdot H = 16.0, 15.8, 15.6, 15.3, 14.8, 14.2$ for $H = 1, 2, 3, 5, 8, 12$, close to the half-plane constant 15.487 of Theorem A. So the threshold is governed by the *first-layer height*, $\mu^* \approx 15.5 h/(H\ell)$, and the factor $c(\theta) \sim \tan \theta$ in any bound $\mu^* \geq c(\theta)\rho$ is genuine. This also matches the paper's $\gamma^* \approx 21$ at $H = \frac{3}{4}$.
- On general coarse meshes only per-mesh certificates hold (§5).

1 Theorem 1: necessity on every (M0) mesh

1.1 Why the v6 approach stalls, and the idea

In `notes/v6/consts` the witness was $\text{curl } I_A F$ with F a fixed smooth profile of width Y . That needs $\text{diam } T \leq \kappa_0 Y$ with $\kappa_0 \ll 1$ on the whole support. On graded meshes $\text{diam } T \sim \text{dist}(T, \Gamma_h)$, so the relative interpolation error is $O(1)$ at every height (Proposition pb:obst). Two observations remove the problem.

- (a) *All the gain sits at the wall, and the wall can be made exact.* For a shear layer $u = q(y)$ with $q(0) = 1$, the half-plane forms per unit length are $B = -q'(0)$, $C = 1$, $A = \int_0^\infty q'^2$. If $\Psi' = q$ is a *polynomial* of degree ≤ 5 on every triangle touching Γ_h , then B and C carry no interpolation error. Under (M0) every such triangle has diameter $\leq D_1 h_\Gamma$, so it suffices that Ψ be quadratic on $\{y \leq y_1\}$ with $y_1 = 2D_1 h_\Gamma$.
- (b) *The unresolved part can be made energetically cheap.* Let q have slope $-s$ up to y_1 , then a small slope $-s\beta$ over a long range, then 0. The gain is $2s$. The energy is $s^2 y_1 \int \omega^2$, which is $O(s\varepsilon)$ with $\varepsilon := sy_1 \rightarrow 0$. Interpolation errors occur only in the two transition bands. There they are $O(1)$ relative to the local energy and hence $O(K^2 s\varepsilon)$, which is negligible against $2s$ once $\varepsilon \lesssim K^{-2}$.

No resolution hypothesis is needed. The only mesh input is the shape-regularity growth bound (Lemma 1).

1.2 Geometric constants

Let $m_\theta = \lceil \pi/\theta \rceil$,

$$D_1 := (\sin \theta)^{-(m_\theta+1)}, \quad c_g := (\sin \theta)^{m_\theta+2}.$$

Lemma 1 (Wall triangles and growth). *[PROVED] Assume shape regularity with minimum angle θ , $|e| \leq h_\Gamma$ for $e \in \mathcal{E}_\Gamma$, and h_Γ small enough that the exterior angle $2\pi/\#\mathcal{E}_\Gamma$ of P_h is $< \theta$.*

(i) *Every $T \in \mathcal{T}_h$ that meets Γ_h has $\text{diam } T \leq D_1 h_\Gamma$.*

(ii) *Every T that does not meet Γ_h (or ∂R) has $\text{dist}(z, \Gamma_h) \geq c_g \text{diam } T$ for each of its vertices z .*

Proof. • *Adjacent triangles.* Two triangles sharing an edge ℓ have diameters in $[\ell, \ell/\sin \theta]$, so their ratio is $\leq 1/\sin \theta$. In a triangle the shortest edge is $\geq \sin \theta$ times the longest (law of sines).

- (i) At a boundary vertex z the angle of Ω_h is $< \pi + \theta$, so z has fewer than $(\pi + \theta)/\theta$ triangles. Any of them is reached from the triangle T_e of a boundary edge $e \ni z$ in at most m_θ edge-adjacent steps, and $\text{diam } T_e \leq |e|/\sin \theta$.
- (ii) Let z be a vertex of T . It is an interior vertex with at most $2\pi/\theta$ triangles, so each is reached from T in at most m_θ steps. Each therefore has diameter $\geq (\sin \theta)^{m_\theta} \text{diam } T$, and its edges at z are $\geq (\sin \theta)^{m_\theta+1} \text{diam } T$. The distance from z to the opposite line is $\geq (\text{an edge at } z) \cdot \sin \theta$. So the star of z contains the open ball $B(z, c_g \text{diam } T)$, and since Ω_h is open that ball misses Γ_h . $\square$

1.3 The witness

Profile.

- $S(t) = 35t^4 - 84t^5 + 70t^6 - 20t^7$ (C^3 smoothstep, $\|S^{(4)}\|_\infty = 840$), extended by 0 for $t < 0$ and 1 for $t > 1$.
- For $\Lambda \geq 1$ put $\beta = 1/\Lambda$ and $T_1 = \Lambda^2$, and define $\omega : [0, \infty) \rightarrow [0, 1]$ by

$$\omega(t) = \begin{cases} 1, & t \leq 1 \\ 1 - (1 - \beta)S(t - 1), & 1 \leq t \leq 2 \\ \beta, & 2 \leq t \leq T_1 \\ \beta(1 - S(t/T_1 - 1)), & T_1 \leq t \leq 2T_1 \\ 0, & t \geq 2T_1. \end{cases}$$

- With $y_1 > 0$, set $q(y) = 1 - s \int_0^y \omega(\tau/y_1) d\tau$ and $\Psi(y) = \int_0^y q$, where s is fixed by $q(\infty) = 0$.
- Explicitly, $\int_0^\infty \omega = \frac{3}{2}(1 + \Lambda - \beta)$ and $\int_0^\infty \omega^2 \leq 2 + 2\beta^2 T_1 = 4$. So

$$\varepsilon := sy_1 = \frac{2}{3(1 + \Lambda - \beta)} \leq \frac{2}{3\Lambda}, \quad \int_0^\infty q'^2 = s^2 y_1 \int \omega^2 \leq 4s\varepsilon.$$

- Ψ is exactly quadratic on $(-\infty, y_1]$ and on $[2y_1, T_1 y_1]$, and constant on $[2T_1 y_1, \infty)$.
- $\Psi \in C^5$, $\Psi^{(5)}$ is Lipschitz, and $\Psi^{(6)} = -s y_1^{-4} \omega^{(4)}(y/y_1)$. So

$$|\Psi^{(6)}| \leq 840 s y_1^{-4} \text{ on the band } [y_1, 2y_1], \quad |\Psi^{(6)}| \leq 840 s \beta (T_1 y_1)^{-4} \text{ on } [T_1 y_1, 2T_1 y_1], \quad \Psi^{(6)} = 0 \text{ elsewhere.}$$

Witness. Let $e_* \in \mathcal{E}_\Gamma$ be any boundary edge. Use coordinates with $e_* \subset \{y = 0\}$ and midpoint 0, so that $P_h \subset \{y \leq 0\}$. Take $X \gg T_1 y_1$ and C^5 cut-offs $\Phi(t) = (1 - t^2)_+^6$ and $\chi \in C^5$ with $\chi = 1$ on $(-\frac{1}{2}, 1]$, $\chi = 0$ on $(-\infty, -1] \cup [2, \infty)$ (the $y < 0$ part only separates the far side of P_h , as in Lemma pb:lem:geom). Then set

$$F(x, y) = \Phi(x/X) \chi(y/X) \Psi(y), \quad v := \text{curl } I_A F,$$

with I_A the global Argyris interpolant. Fix

$$y_1 := 2D_1 h_\Gamma, \quad K := E_2(\theta) (3/c_g)^4 \cdot 840, \quad K_1 := 2 + K \sqrt{40/c_g}, \quad \Lambda := \max(1, K_1^2).$$

Here $E_2(\theta)$ is the explicit Argyris constant of Lemma pb:lem:arg ($E_2 = 179, 39.0, 17.3$ at $\theta = 20^\circ, 30^\circ, 45^\circ$).

Theorem 2 (Penalty necessity on every (M0) mesh). *[PROVED] Assume (M0) with minimum angle θ and constant c_0 . There is $h_*(\theta, c_0, L) > 0$ such that for $h \leq h_*$ and suitable X (below), $v \in Z_h$ for every $k \geq 4$ and*

$$\frac{2B(v) - A(v)}{C(v)} \geq \frac{s}{2} = \frac{\varepsilon}{4D_1 h_\Gamma}, \quad \text{hence} \quad \mu^* \geq \frac{\varepsilon}{4D_1} \frac{h}{h_\Gamma} \geq c_* \rho, \quad c_* := \frac{c_0}{6D_1(1 + \Lambda)}.$$

Neither (M1) nor (M2) is used, nor any quasi-uniformity or star or layer structure. e_ may be any boundary edge.*

Proof. All statements are per unit length of $\{|x| < X\}$ after dividing by $\int \Phi^2$. " $o(1)$ " denotes terms that vanish as, in order, $X/(T_1 y_1) \rightarrow \infty$ and then $X \rightarrow 0$ (i.e. $h_\Gamma \rightarrow 0$ with X/h_Γ fixed). Both limits are with θ, c_0 fixed, and each $o(1)$ term has an explicit bound of the type written, which fixes h_* .

1. *Membership.*

- $F \in C^5$ with bounded, piecewise continuous $D^6 F$, so $I_A F$ is a globally C^1 piecewise P_5 function and v is continuous, piecewise P_4 and divergence-free.
- $\text{supp } F \cap \overline{\Omega_h} \subset \{|x| \leq X, \gamma(x) \leq y \leq 2X\}$.
- By Lemma 1(ii), the triangles meeting it have diameter $\leq (2X + \delta)/c_g$. So v vanishes on ∂R once $3X/c_g < L - 1$.

2. *The wall is exact.*

- Let δ, σ be the depth and slope bounds of Lemma pb:lem:geom on $|x| \leq X'$; they are $O(X^2)$ and $O(X)$.
- Every triangle meeting Γ_h lies in $\{-\delta \leq y \leq D_1 h_\Gamma = y_1/2\}$ (Lemma 1(i)). There $F = \Phi(x/X)\Psi(y)$ with $\Psi(y) = y - sy^2/2$, so $D^6 F = \sum_{i \geq 4} \binom{6}{i} \Phi^{(i)} X^{-i} \Psi^{(6-i)}$.
- Lemma pb:lem:arg then gives $|D^j(F - I_A F)| \leq E_j(D_1 h_\Gamma)^{6-j} C_\Phi(sX^{-4} + X^{-5} + y_1 X^{-6})$ there, which is $o(1) s h_\Gamma^{1-j}$ for $j = 1, 2$.
- Hence $B(v) = B_{\Gamma_h}(w) + o(1)s$ and $C(v) = C_{\Gamma_h}(w)(1 + o(1))$ with $w = \text{curl } F$.
- As in step 2 of Theorem pb:thmA, using $nds = (\gamma', -1)dx$, $q(\gamma) = 1 + O(s\delta)$ and $\Psi(\gamma) = O(\delta)$, one gets $B_{\Gamma_h}(w) = s(1 + O(s\delta) + O(\sigma/(sX)))$ and $C_{\Gamma_h}(w) = 1 + O(s\delta + \sigma^2)$.
- Since $s\delta \sim \varepsilon X^2/h_\Gamma$ and $\sigma/(sX) \sim h_\Gamma/\varepsilon$, both are $o(1)$ as $h_\Gamma \rightarrow 0$.

3. *Band errors.* Let T meet $[y_1, 2y_1]$.

- T does not meet Γ_h , because $y_1 > D_1 h_\Gamma$.
- By Lemma 1(ii) its lowest vertex has $y \geq c_g \text{diam } T - \delta$, and that vertex has $y \leq 2y_1$. So $\text{diam } T \leq 3y_1/c_g$, and $T \subset \{y \leq 2y_1 + 3y_1/c_g \leq 5y_1/c_g\}$.
- Lemma pb:lem:arg gives $|D^2(F - I_A F)| \leq E_2(3y_1/c_g)^4 \cdot 840 s y_1^{-4}(1 + o(1)) = Ks(1 + o(1))$ on T .
- The same holds on the second band with $y_1 \rightarrow T_1 y_1$ and $s \rightarrow s\beta$. No triangle meets both bands, because $T_1 \geq 2 + 3/c_g$.
- On all other triangles Ψ is a polynomial of degree ≤ 2 , so the error is $o(1)$ as in step 2. On the χ -region $y \in [X, 2X]$, triangles have $\text{diam} \leq 2X/c_g + o(1)$, giving an error energy $O(K^2 \Psi_\infty^2/X^2) = o(1)$.
- With $A(\text{curl } z) \leq 4\|D^2 z\|^2$:

$$A(\text{curl}(F - I_A F)) \leq 4K^2 s^2 \left(\frac{5y_1}{c_g} + \beta^2 \frac{5T_1 y_1}{c_g} \right) + o(1) = \frac{40K^2}{c_g} s\varepsilon + o(1) \quad (\beta^2 T_1 = 1).$$

4. *Energy of w .*

- The separable half-plane identity (integrating by parts in x and y , with $\Psi(0) = 0$) gives

$$A(w) = \int \Phi'^2 \cdot 2 \int G'^2 + \int \Phi^2 \int G''^2 + \int \Phi'^2 \int G^2, \quad G = \chi \Psi.$$

- Per unit length this is $\int q'^2 + O(T_1 y_1/X^2) + O(\Psi_\infty^2/X^3) = \int q'^2 + o(1) \leq 4s\varepsilon + o(1)$.
- The sliver $\{\gamma < y < 0\}$ adds $O(s^2 X^2) = o(1)s$.

5. *Budget.*

- Minkowski gives $A(v)^{1/2} \leq (4s\varepsilon)^{1/2} + (40K^2s\varepsilon/c_g)^{1/2} + o(1)$, i.e. $A(v) \leq s\varepsilon K_1^2 + o(1)$.
- Therefore $2B - A \geq 2s - s\varepsilon K_1^2 - o(1)$ $s \geq s(1 - o(1))$, because $\varepsilon \leq 2/(3\Lambda) \leq 1/K_1^2$.
- With $C = 1 + o(1)$, we get $(2B - A)/C \geq s/2$ for $X/(T_1 y_1)$ large and h_Γ small.
- Then $N_h(v, v) < 0$ whenever $\mu/h < s/2 = \varepsilon/(4D_1 h_\Gamma)$.
- (M0) gives $h/h_\Gamma \geq c_0 \rho$. □

Remark 3 (Size of the constants). In $\log_{10}$, $c_*/c_0 \approx -67, -39, -23$ (formula of Theorem 2) at $\theta = 20^\circ, 30^\circ, 45^\circ$. The support height $2T_1 y_1$ is $\approx 10^{75} h_\Gamma$ at 30° , so h_* is correspondingly tiny. The losses are the worst-case growth factor c_g^{-4} (sharp only for pathological stars) and the Taylor-based E_2 ($\approx 7 \cdot 10^4$ pessimistic). Both enter squared through Λ . The theorem settles the *scaling* question listed as open in paper 14_open.tex. It does not give a practical threshold. On real meshes the same witness family gives modest, positive quotients without any two-slope spreading (§2).

Remark 4 (Clough–Tocher). [**PROVED modulo the standard HCT estimate** $|f - I_{HCT} f|_{W^{2,\infty}(T)} \leq C_{HCT}(\theta) h_T^2 |f|_{W^{4,\infty}(T)}$] Replace I_A by the HCT interpolant on the macro mesh. Ψ quadratic is reproduced, the band bound becomes $C_{HCT}(3y_1/c_g)^2 \|\Psi^{(4)}\|$ with $|\Psi^{(4)}| \leq s \|S''\| y_1^{-2}$ on band 1, and every step goes through. v is then continuous, piecewise P_2 on the CT refinement and divergence-free, so the result holds for every $k \geq 2$.

Remark 5 (Relation to the v6 routes and to bounded patches). The report notes/v6/consts proposed two routes: (i) a sharp computer-assisted interpolation bound, and (ii) half-plane compactness (Theorem B there). Neither is needed. The witness is built so that the only unresolved region carries a vanishing share of the energy.

- *Bounded patches do suffice.* The support has size $\leq C(\theta, c_0) h_\Gamma$, hence bounded graph radius $R(\theta, c_0)$. So the bounded-patch approach succeeds, with this explicit witness in place of a compactness argument.
- *What fails is small R .* Stars and edge patches fail (Proposition pb:counter). The H -scan of §4 and the window data indicate that R must grow as $\theta \rightarrow 0$: windows must span $\gtrsim H$ edges.

2 Numerical companion: the same witness on graded and column meshes

[NUMERICAL] (layer_witness.py, .log).

- *Setup.* Half-plane meshes with $h_\Gamma = 1$ and $v = \text{curl } I_A F$ as above. For the numerics ω is built from the C^5 smoothstep, and β, T_1 are small ($\beta = 0.3, T_1 = 4$) or trivial ($\beta = 1, T_1 = 1$: one slope). The quantity reported is $Q = h_\Gamma(2B - A)/C$.
- *“Continuum”.* This is the separable half-plane value, which the discrete value must approach where the mesh resolves F .

mesh (min angle)	(y_1, β, T_1)	continuum ($X = 800$)	discrete $X = 200$	discrete $X = 800$
columns $H = 5$ (11.3°)	(5, 0.3, 4)	0.0692	0.0654	0.0693
	(5, 1, 1)	0.1372	0.1367	0.1377
columns $H = 8$ (7.1°)	(8, 0.3, 4)	0.0049	−0.0028	0.0050
	(8, 1, 1)	0.0812	0.0795	0.0815
dyadic (26.6°)	(1, 0.3, 4)	0.4737	0.4792	0.4796
	(1, 1, 1)	0.7015	0.7326	0.7327
columns $H = 5$ + dyadic above (11.3°)	(5, 0.3, 4)	0.0692	0.0807	0.0876

The witness is positive on the column meshes, where every boundary star fails, and on the graded layer. The cost of the χ cut-off (Ψ_∞^2/X_c^3) explains the small $H = 8$, $\beta = 0.3$ value. In the numerics the one-slope profile is better: these meshes are not adversarial enough for the two-slope device to pay off. The quotients are $O(1/y_1)$ and far below the true thresholds (next section). This witness is a proof device, not an optimiser.

3 Stacked column meshes: an exact witness for arbitrarily tall boundary triangles

Proposition 6. *[PROVED, exact rational arithmetic] Let the half-plane mesh consist of cells $[i, i+1] \times [jH, (j+1)H]$ ($H > 0$ real) split by the forward diagonal. Its boundary triangles are right-angled with apex angle $\arctan(1/H)$, and every boundary star is the counterexample star $\mathcal{S}^{(H)}$.*

- Let Φ be the uniform cubic B-spline with integer knot spacing $d \geq 4H$ (knots on vertical mesh lines).
- Let G be the C^1 piecewise quadratic with knots at $H, 3H, 5H$: $G = y - y^2/(2H)$ on $[0, H]$, then $G'' = -c$ on $[H, 3H]$ and $+c$ on $[3H, 5H]$, with $c = 1/(8H)$, and $G = 0$ after.

Then $F = \Phi(x)G(y)$ is C^1 and piecewise P_5 on the mesh, so $v = \text{curl } F \in Z_h$ for all $k \geq 4$ exactly, with no interpolation. With $D = d/H$,

$$H \cdot \frac{h_\Gamma(2B - A)}{C} = \frac{2265D^4 - 2800D^2 - 6944}{2416D^4} \geq 0.8538 \quad (D \geq 4).$$

Hence $\mu^* \geq 0.85 h/(Hh_\Gamma)$ on any mesh containing this configuration over the witness support $(4d \times 5H)$, at any mesh size.

Proof. • *Forms.* For $F = \Phi G$ with $G(0) = 0$: $B = \frac{1}{H} \int \Phi^2$, $C = \int \Phi^2$ and $A = 2 \int \Phi'^2 \int G'^2 + \int \Phi^2 \int G''^2 + \int \Phi'^2 \int G^2$ (integration by parts, compact support). The spline and G integrals are rational: $\int \Phi^2 = \frac{151d}{315}$, $\int \Phi'^2 = \frac{2}{3d}$, $\int \Phi''^2 = \frac{8}{3d^3}$, $\int G^2 = \frac{31H^3}{60}$, $\int G'^2 = \frac{5H}{12}$, $\int G''^2 = \frac{17}{16H}$.

- *Monotonicity.* The expression is increasing in D .
- *Checks.* `exact_columns.py` (SymPy, exact) evaluates the expression. For example $Q = 16503/96640$ at $H = 5$, $d = 20$. It is cross-checked by direct Gauss quadrature of $\int 4F_{xy}^2 + (F_{yy} - F_{xx})^2$ on the cells (agreement to 10^{-16}). $\square$

The limit as $D \rightarrow \infty$ is $(1 - \frac{1}{2M^3})/H$, with $M = 2$ here; $M = 3$ gives $0.9815/H$. Exactness needs the layer lines parallel to Γ_h , so on the circle this is a flat-model statement. Under a fixed perturbation (non-explicit, as in Proposition pb:counter) it transfers to the polar column mesh for large N .

4 Goal 2: a global (M0)–(M2) mesh containing the star counterexample

[NUMERICAL] (`global_mesh.py`; logs `global_mesh_H5_N128.log`, `_H5_N256`, `_H8_N128`, `_Hscan`).

Construction.

- P_h is the regular N -gon, with chord $\ell = 2 \sin(\pi/N)$.
- *Layer 0.* Radial columns $[z_i, z_{i+1}, r_1 z_{i+1}, r_1 z_i]$ with $r_1 = 1 + H\ell$, split on the forward diagonal. The star of every boundary vertex is a perturbation of $\mathcal{S}^{(H)}$, of relative size $H \cdot 2\pi/N$ (column widening).
- *Rings 1..J.* Each cell sits over two cells of the previous ring, has radial thickness equal to its width, and is cut into 3 triangles from the hanging bottom midpoint.
- *Outer part.* $N/2^J$ rays to ∂R , with m structured layers on the forward diagonal.

Checked in code for every mesh: (M0) with $c_0 = 1$; no singular vertex (M1); Θ_0 of (M2); minimum angle.

Results ($H = 5$). Patch thresholds $\lambda = (\mu/h)^* h_\Gamma$ for fields supported in the patch (vanishing on its non- Γ_h boundary), and the global threshold:

	1 star	2 stars	3	4	8	16	layer 0	global γ^*
$N = 128$ ($\Theta_0 = 0.83, 11.25^\circ$)	none	0.432	1.110	1.459	2.096	2.624	3.053	3.0562
$N = 256$ ($\Theta_0 = 0.77, 11.28^\circ$)	none	0.497	1.177	1.520	2.121	2.641	3.127	3.1286

("none": no negative direction at $\mu/h \geq 10^{-3}/h$.)

- *Consistency with the v6 certificates.* The single star fails and three stars give $\lambda > 1$. This is the $\mathcal{S}^{(5)}$ statement of Proposition pb:counter, reproduced on the actual global mesh.
- *Scaling with ρ .* The global threshold is attained essentially in layer 0. Varying the box at fixed h_Γ :

L	$N = 128$			$N = 256$		
	ρ	μ^*	μ^*/ρ	ρ	μ^*	μ^*/ρ
2.5	21.57	65.94	3.0562	23.02	72.01	3.1286
5	67.41	206.02	3.0560	135.54	424.04	3.1284
10	207.68	634.67	3.0560	415.93	1301.22	3.1285

So $\mu^* = \gamma^* \rho$ with γ^* independent of L to four digits: *the star counterexample is local and the global threshold scales exactly like ρ .*

- $H = 8$ (7.1°): no witness on 1, 2 or 3 stars; 4 stars give 0.1375 (the flat value is $> \frac{1}{4}$; the difference is the column widening $H \cdot 2\pi/N = 0.39$); $\gamma^* = 1.8527$ for $L = 2.5, 5, 10$.
- Theorem 2 makes the ρ -scaling a theorem for this family as $h \rightarrow 0$.

Dependence on the column height ($N = 128, L = 2.5$, global threshold):

H	1	2	3	5	8	12
min angle	12.96°	14.53°	14.05°	11.25°	7.10°	4.75°
γ^*	15.999	7.884	5.199	3.056	1.853	1.185
$\gamma^* H$	16.0	15.8	15.6	15.3	14.8	14.2

$\gamma^* H$ stays near the half-plane constant 15.487 of the smooth witness at scale $Y =$ layer height, and the paper's production meshes (layer height $\frac{3}{4}$, $\gamma^* \rightarrow 21.4$) give $\gamma^* H \approx 16$. [**NUMERICAL, conjecture**]: $\mu^* \approx (15\text{--}16) h/\delta_1$, with δ_1 the height of the first layer, is the sharp form of the necessity. The $\tan \theta$ -type decay of $c(\theta)$ in Theorem 2 is therefore genuine, although its size there is not.

Code note. `code/penalty.py` caches the factorisation data in a default argument keyed by `id(S)`. A patch `Space` built inside a function and freed can hand its `id` to the next one, and then stale data are reused (it happened here once: 2.62 instead of 3.05 for layer 0). `global_mesh.py` clears the cache before each call. `lamstar_family.mesh_patches` reassigns `S` in a loop, so the old object is alive when the new one is created and the paper’s tables are not affected.

5 Goal 3: tall boundary triangles on coarse meshes

What holds, for $\mu^* \gtrsim \rho$ when boundary triangles are tall (apex $< 30^\circ$, so $\lambda_T < 0$ and Corollary pb:rho does not apply):

- (a) **[PROVED]** *Asymptotically, for all shapes* (Theorem 2): $\mu^* \geq c_*(\theta, c_0)\rho$ for $h \leq h_*$. Tallness enters only through θ . Because h_* is astronomically small, this says nothing about a given coarse mesh.
- (b) **[PROVED, exact, every mesh size]** *Layered (column) boundary layers of any height* (Proposition 6): $\mu^* \geq 0.85 h/(Hh_\Gamma)$ in the flat model. This is exactly the tall-right-triangle class of Remark pb:limits and the class of the star counterexamples.
- (c) **[PROVED, per mesh]** *Any given coarse mesh:*
 - Necessity is certified by a finite computation: an exact or interval witness on a first-layer window, as in Proposition pb:counter and the $W = 4$ certificate of the paper.
 - Stars and edge patches are *not* enough (Proposition pb:counter). Windows spanning $\gtrsim H$ edges are.
 - On the polar column meshes the window of W stars gives $\lambda > 0$ from $W = 2$ ($H = 5$) or $W = 4$ ($H = 8$), and layer 0 alone reproduces the global threshold to 10^{-3} .
- (d) **[NUMERICAL]** *Size:* $\gamma^* \approx 15.5/H$ on column layers (§4). No mesh-independent $c > 0$ exists without θ .
- (e) **[OPEN]** An explicit, moderate constant for general (M0)–(M2) coarse meshes (e.g. $\mu^* \geq c h/\delta_1$ with $c \approx 1$). This would need either a sharp interpolation estimate for the witness on unresolved cells, or an exact discrete layer witness on general (non-tensor) first layers.

6 Suggested changes to the paper (not made)

- In `14_open.tex`, move "Penalty necessity for tall boundary triangles ... and a global mesh realising the counterexamples" from Open to:
 - Proved: Theorem 2, qualitative in the constant, plus the exact column-layer Proposition 6;
 - Numerical: the global mesh of §4;
 - Open: keep the explicit-moderate-constant question of (e) above.
- In `08_penalty.tex`, after Corollary pb:corA, replace "A uniform necessity statement under (M0)–(M2) alone remains open" by Theorem 2. Replace "we have not written it out" (global mesh) by a pointer to §4.

Files (all in notes/v7/penalty/)

- `layer_witness.py`, `.log`: the Theorem 2 witness on half-plane meshes (≈ 75 s).
- `exact_columns.py`, `.log`, `exact_columns_closedform.log`: Proposition 6 (< 30 s).
- `global_mesh.py` and `global_mesh_*.log`: the global mesh, its (M0)–(M2) checks, and the patch and global thresholds (15–45 s per run; usage `python3 global_mesh.py H N [all|star|g]`).